



Lot 3: Découverte

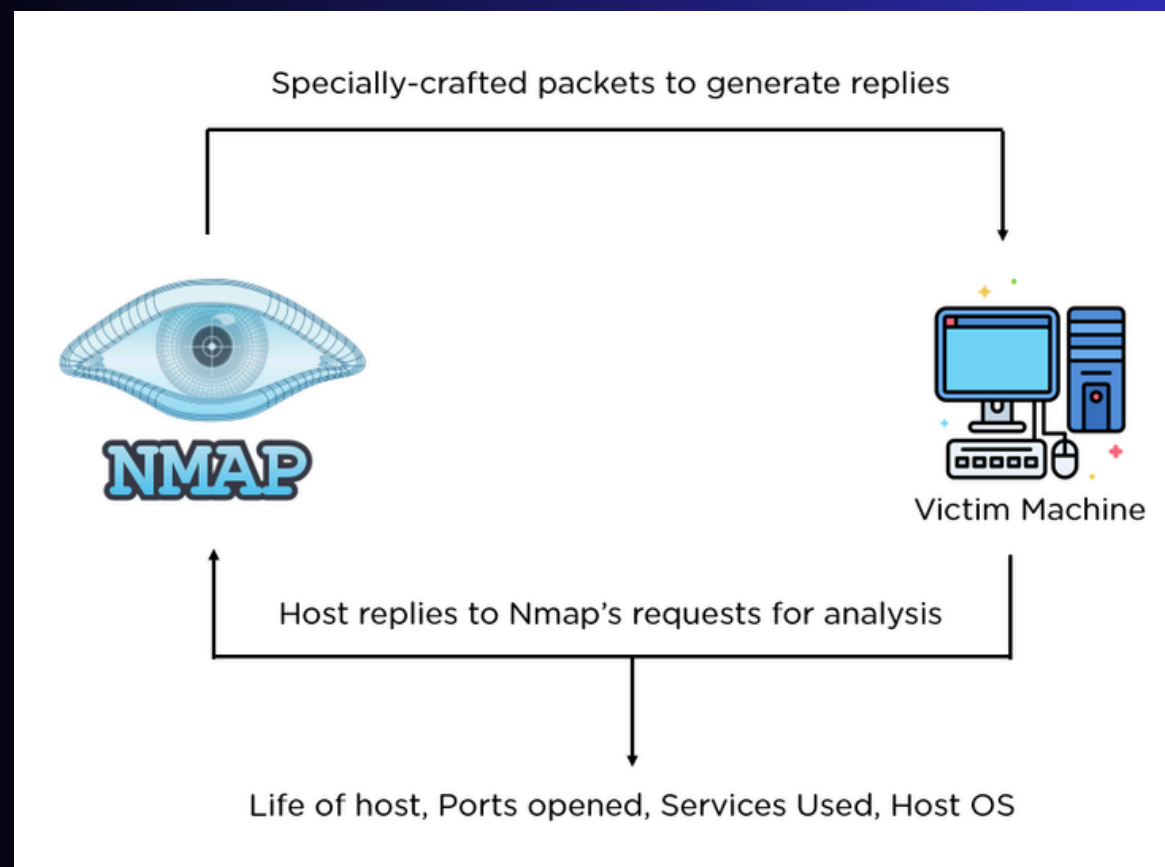
Découvrir les machines du réseau

Découvrir les OS des machines du réseaux (fingerprint)

Découvrir les services disponibles sur les machines distantes

Découvrir les versions des services disponibles

Découvrir les machines du réseau



`sudo nmap -sn <adresse réseau destination>`

```
adminetu@RT007:~$ sudo nmap -sn 192.168.111.117/24
[sudo] Mot de passe de adminetu :
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:32 CEST
Nmap scan report for 192.168.111.1
Host is up (0.00044s latency).
MAC Address: 84:A0:6E:98:5B:BD (Sagemcom Broadband SAS)
Nmap scan report for RTP21 (192.168.111.68)
Host is up (0.00033s latency).
MAC Address: 84:69:93:0A:01:C4 (HP)
Nmap scan report for 192.168.111.71
Host is up (0.00020s latency).
MAC Address: 84:69:93:0A:01:AC (HP)
Nmap scan report for RTP35 (192.168.111.72)
Host is up (0.00030s latency).
MAC Address: 84:69:93:07:A3:51 (HP)
Nmap scan report for 192.168.111.73
Host is up (0.00026s latency).
MAC Address: 84:69:93:07:A4:57 (HP)
Nmap scan report for RTP34 (192.168.111.77)
Host is up (0.00031s latency).
MAC Address: 84:69:93:07:A3:6B (HP)
Nmap scan report for RTP23 (192.168.111.80)
Host is up (0.00035s latency).
MAC Address: 84:69:93:06:51:F1 (HP)
```

Découvrir les OS des machines du réseau (fingerprint)

```
sudo nmap -O <votre_adresse_ip>/<masque_de_sous_réseau>
```

```
adminetu@RT007:~$ sudo nmap -O 192.168.111.117/24
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:40 CEST
Nmap scan report for 192.168.111.1
Host is up (0.00085s latency).
Not shown: 845 closed tcp ports (reset), 148 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
5060/tcp  open  sip
9000/tcp  open  cslistener
MAC Address: 84:A0:6E:98:5B:BD (Sagemcom Broadband SAS)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.93%E=4%D=10/17%OT=53%CT=135%CU=42097%PV=Y%DS=1%DC=D%G=Y%M=84A06
OS:E%TM=68F20121%P=x86_64-pc-linux-gnu)SEQ(SP=C8%GCD=1%ISR=CE%TI=Z%CI=Z%II=
OS:I%TS=U)OPS(O1=M5B4NNSNW4%O2=M5B4NNSNW4%O3=M5B4NW4%O4=M5B4NNSNW4%O5=M5B4N
OS:NSNW4%O6=M5B4NNS)WIN(W1=16D0%W2=16D0%W3=16D0%W4=16D0%W5=16D0%W6=16D0)ECN
OS:(R=Y%DF=Y%T=40%W=16D0%O=M5B4NNSNW4%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+F=A
OS:S%RD=0%Q=)T2(R=N)T3(R=Y%DF=Y%T=40%W=16D0%S=0%A=S+F=AS%O=M5B4NNSNW4%RD=0
OS:%Q=)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z
OS:%A=S+F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=N
OS:)%U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%
OS:DFI=N%T=40%CD=S)

Network Distance: 1 hop

Nmap scan report for RTP21 (192.168.111.68)
Host is up (0.00057s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
902/tcp   open  iss-realsure
MAC Address: 84:69:93:0A:01:C4 (HP)
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.3
```

Découvrir les services disponibles sur les machines distantes

```
sudo nmap -sV <adresse_ip_cible>
```

```
adminetu@RT007:~$ sudo nmap -sV 192.168.111.127
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:46 CEST
Nmap scan report for 192.168.111.127
Host is up (0.00017s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
MAC Address: 84:69:93:07:A4:8A (HP)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds
```

Découvrir les versions des services disponibles

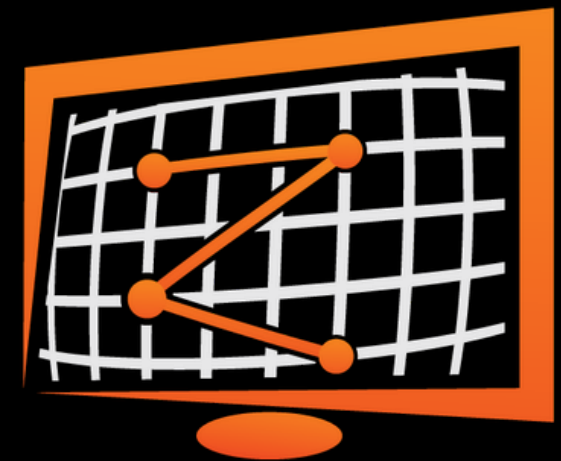
```
sudo nmap -sV --version-all <adresse_ip_cible>
```

```
adminetu@RT007:~$ sudo nmap -sV --version-all 192.168.111.127
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-17 10:50 CEST
Nmap scan report for 192.168.111.127
Host is up (0.00042s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
22/tcp    open  ssh              OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
MAC Address: 84:69:93:07:A4:8A (HP)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```


Autres outils de scan de réseau

- netdiscover - massscan -
zmap - unicorn scan - rustscan



Les risques de nmap et du scan de réseaux

Pourquoi ?

Peut être considéré comme du scan actif
(répréhensible par la loi)

Surcharge de la bande passante

Que faire ?

Demander l'autorisation aux administrateurs réseaux

Prévenir les administrateurs réseaux quand on démarre un scan
(peut encombrer la bande passante)

Utiliser l'option -T2 (Polite)
(Pour ne pas saturer le réseau)

Source

- nmap.org



Conclusion

Risque légaux

scanne réseau

plusieurs outils